

Skills Summary

Where Limit Work Goes Wrong

Use this reference while completing your worksheet or when studying.

Skill 1 — Is Direct Substitution Even Legal?

The test: substitute the target into the *denominator* first. If it is not zero, the function is continuous there and the limit is the value — substitute and stop. If it is zero and the numerator is not, you have a vertical asymptote. If both are zero you have $\frac{0}{0}$, which is an **indeterminate form**: it is not an answer, it is a message saying “rewrite me”. Factor and cancel, or multiply by the conjugate.

Example: Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Step 1. Substituting 3 gives $\frac{0}{0}$, so substitution has failed — that is a signal to rewrite, never a value.

Step 2. $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$, and $3 + 3 = 6$.
 $\Rightarrow 6$

Try it: Evaluate $\lim_{x \rightarrow 7} \frac{x^2 - 49}{x - 7}$

check: 14

Skill 2 — The Sign of the Denominator Decides

The test: at a vertical asymptote the size is always infinite, so the only question is the *direction*. Read two signs, not one: the sign the numerator is heading to, and the sign of the denominator *on the side you are approaching from*. Same signs give $+\infty$; opposite signs give $-\infty$. An even power in the denominator is positive on both sides.

Example: For $F(x) = \frac{5}{x - 1}$ find the limit from each side of 1.

Step 1. The numerator is a fixed positive 5, so both answers are decided by the sign of $x - 1$ on each side.

Step 2. Left of 1: $x - 1$ is a small negative, so 5 over it is large negative. Right of 1: $x - 1$ is a small positive, so it is large positive.

$\Rightarrow$ left: $-\infty$ right: $+\infty$

Try it: For $G(x) = \frac{3}{x + 4}$ find the limit from each side of -4 .

check: left $-\infty$, right $+\infty$

Skill 3 — Diagnosing Someone Else's Work

How to write a find-and-fix answer: a complete answer has two parts, and both are graded. **Fix:** redo the limit correctly and state the value. **Diagnose:** name the specific move that was illegal — “treated $\frac{0}{0}$ as a value”, “cancelled across a + instead of a factor”, “never checked the sign of the denominator”. “It is wrong” is not a diagnosis.

Example: A student says $\lim_{x \rightarrow 3^+} \frac{x-8}{x-3} = +\infty$ “because it blows up”. Fix it.

Step 1. The size is right but the direction was never argued, so check both signs on the right of 3.

Step 2. Numerator near $3-8 = -5$, negative; denominator $x-3$ a small positive. Negative over positive is large negative, so the direction is downward.

$\Rightarrow -\infty$ — the diagnosis is that only the denominator was looked at.

Try it: A student says $\lim_{x \rightarrow 1^-} \frac{x-9}{x-1} = -\infty$ “because the top is negative”. Fix it.

$\infty+$ check